

SlimIoT Pilot Management Platform

Product Requirements Document (PRD) v1.0

Purpose

Centralized platform for planning, deploying, monitoring and reporting pilot studies across hospitals, eldercare, wellness and research environments.

Business Goals

Reduce pilot administration effort, improve compliance, automate reporting, generate evidence, and scale across multiple pilot sites.

User Types

Administrator, Pilot Manager, Site Coordinator, Viewer.

Navigation

Dashboard → Organizations → Sites → Pilots → Participants → Devices → Analytics → Reports → Administration

Screen-by-Screen Wireframes

Pilot Command Center

Active Pilots | Participants | Devices | Capture Rate
Pilot List
Alerts & Notifications

Pilot Detail

Overview | Participants | Devices | Analytics | Reports

Participant Management

Participant ID | Category | Status | Compliance

Device Management

Device ID | Type | Status | Last Sync

Analytics Dashboard

Sleep Duration | Efficiency | HR | Compliance Trends

Feedback Dashboard

Comfort | Ease of Use | Recommendation

Report Generator

Pilot | Date Range | Format | Generate

Database Schema

```
Organization(id, name, industry, address, contact_person)
Site(id, organization_id, name, city, state)
Pilot(id, site_id, name, start_date, end_date, product_type, status)
Participant(id, pilot_id, participant_code, category, gender, age_band, consent_status)
Device(id, serial_number, device_type, firmware_version, status)
DeviceAssignment(id, participant_id, device_id, assigned_on, returned_on)
Observation(id, participant_id, timestamp, metric_type, metric_value)
Feedback(id, participant_id, comfort_score, usability_score, recommend, comments)
```

Recommended Data Hierarchy

Organization → Site → Pilot → Participant → Device → Observation

API Requirements

POST /api/auth/login
POST /api/auth/logout

GET/POST/PUT/DELETE /api/pilots
GET/POST/PUT /api/participants
GET/POST/PUT /api/devices
GET/POST /api/device-assignments
GET/POST /api/observations

GET /api/reports/pilot
GET /api/reports/device
GET /api/reports/participant

User Permissions Matrix

Admin: Full access. Pilot Manager: Pilot operations. Coordinator: Site-level management. Viewer: Read-only access.

Reporting Requirements

- Operational Report
- Participant Report
- Device Report
- Executive Summary Report
- PDF, Excel and CSV exports

Future AI & Analytics Roadmap

- Automated anomaly detection
- Pilot Intelligence Assistant
- Cross-pilot benchmarking
- Predictive compliance modelling
- Evidence Repository